

Supplemental Instructions for 915 MHz Meshtastic Yagi Antenna Build

While the basic instruction set for this build is complete as far as it goes, it does not really go far enough. This is due to the fact that the instruction set was developed and written as applicable to the assembly of the antenna from the kits that were provided to the GCARC Club members involved in the original build event. Those kits included pre-cut and finished element rods for the antenna. Thus, that instruction set did not reference cutting and finishing the element rods in any manner whatsoever, and in fact, the instructions never actually identified the specific lengths of the elements. While that data can be gleaned from a careful read of the YagiCad® screen capture provided, which shows the element lengths and positions, it was decided to develop and provide these specific element preparation instructions.

Based on the lengths of the individual element rods, a total of three of the twelve-inch brass rod stock recommended for this build will be needed if constructing a single antenna. If constructing multiple antennas (or preparing kits for such a build), it is possible to use less than three full rods for a single antenna, resulting in less waste rod stock. We will discuss this concept more in depth later on.

Table 1 - Element lengths and positions

Element ID	Element Name	Element Location	Element Length	Qty
REF	Reflector	0.00"	6.28125" (6-9/32")	1
DE*	Driven Element	2.59375" (2-19/32")	3.03125" (3-1/32")	2
DE*	Driven Element	2.59375" (2-19/32")	3.03125" (3-1/32")	
D1	Director #1	3.5625" (3-9/16")	5.65625" (5-21/32")	1
D2	Director #2	5.875" (5-7/8")	5.59375" (5-19/32")	1
D3	Director #3	8.65625" (8-21/32")	5.53125" (5-17/32")	1

**DE is duplicated due to the split nature of the driven element. Two identical element rods are used, located at the same relative distance from the reflector element, one on either side of the boom and with an insulator between them.*

Table 1 shows the element lengths and positions on the boom. The boom element locations are pre-indicated during the 3-D print of the boom, so these locations are provided purely for reference purposes – you will not need to use these numbers.

The preparation of the element rods requires the use of a strong cutting pliers, preferably end-cutting pliers. In addition, a means must be available to square off the end of the element rod where the rod was cut. This is best done with a Dremel® tool with a cut-off wheel in its tool collet. While it can also be done with a good file, that is a very tedious and dirty job. An ultrafine-point Sharpie® marker will be needed for marking the rod stock during measurement, and an accurate means of measuring to the nearest thirty-second

of an inch is needed. I recommend a steel rule such as the *Mr. Pen Steel Rulers* offered by Amazon (<https://www.amazon.com/gp/product/B07ZS4JSQP>) for a very reasonable price. These rulers have the advantage of clear and accurate markings with no offset of the “zero” position. This means that you can simply line up one end of the rod stock with the end of the ruler and go from there.

Once all of the required equipment is at hand, start by marking out the element lengths on the rod stock as follows:

- on the first rod, measure and mark $6\text{-}9/32$ ” from one end of the rod, and then measure and mark $5\text{-}17/32$ ” from the opposite end of the rod, and set that rod aside;
- on the second rod, measure and mark $5\text{-}21/32$ ” from one end of the rod, and then measure and mark $5\text{-}19/32$ ” from the opposite end of the rod, and set that rod aside; and
- on the third rod, measure and mark $3\text{-}1/32$ ” from one end of the rod, and then measure and mark $3\text{-}1/32$ ” from the opposite end of the rod, and set that rod aside.

The next step is to rough-cut the element rods, based upon the marks that were just made. If the above instructions were followed properly, each rough-cut rod will end up having one square finished end after the rough cuts are made.

Using the end-cutting pliers, cut each piece of rod stock about a thirty-second of an inch away from the measured mark **towards the center of the rod.** This means that you will be making the first cut between the two marks on the rod stock, and then the second cut will be made between the remaining measured mark and the cut end of the rod stock. Once all three pieces of rod stock have been cut near each of the measured marks, the yield will be six element rods which will each have one finished end, one rough end, and each will be about a thirty-second of an inch longer than its measured and marked length.

Finish each element rod by grinding or filing off the rough-cut end, bringing the cut to a clean square finish and the element rod to its specified finished length. Lay the rod against the ruler periodically as you grind or file it to ensure compliance with the length requirement for the element rod on which you are working at that time.

Finally, dress off any flash that may be “mushrooming” the cut end of the rod to give a nice clean and round element rod end that will fit properly into the hole on the boom. This will complete the preparation of the element rods from three twelve-inch lengths of brass rod stock, for a single antenna construction project.

If you are in the position of needing to prepare element rods for multiple antennas to be built, for example, if you are preparing kits for future use in building the antenna in quantity, some reduction in rod stock waste can be achieved. In this case, you would be laying out and cutting the element rods for three antennas using only eight lengths of rod stock instead of nine lengths as would be used in the single-antenna process.

Begin by measuring and marking three sets of reflector and director (or *parasitic*) element rods in the same manner as was described in the single-antenna procedure. That means that you will be measuring and marking out the element lengths on the rod stock as follows:

- on the first rod, measure and mark $6\text{-}9/32''$ from one end of the rod, and then measure and mark $5\text{-}17/32''$ from the opposite end of the rod, and set that rod aside; and
- on the second rod, measure and mark $5\text{-}21/32''$ from one end of the rod, and then measure and mark $5\text{-}19/32''$ from the opposite end of the rod, and set that rod aside.

Repeat that measure and mark sequence to get three sets of rods, which yields four elements each for three antennas. Rough-cut and then finish these element rods as described previously.

Next, measure and mark two lengths of rod stock at four-inch intervals, effectively marking the rod stock in thirds. Cut these rod segments at each of the marks, which will give you six four-inch rod segments. Four of those rod segments will be finished at one end, while the remaining two rod segments – the ones cut from the centers of the rod stock lengths – will be rough-cut on both ends. Using the Dremel® tool or the file, dress and finish one end of each of the two rods that have both ends cut.

Continue by measuring and marking, **working from the finished end of the rod segments**, $3\text{-}1/32''$ from the finished end of each of the six four-inch rod segments. Then, rough-cut those rod segments at about $1/32''$ past the mark towards the cut end of the rod segment. Finally, using the tool of your choice, grind or file the rod segments to their finished element rod length of $3\text{-}1/32''$ as was done with the parasitic elements earlier.

This completes the element rod preparation procedure when multiple antennas are being prepped, or when kits are being assembled. Note that a standard drinking straw makes a great storage tube for the finished element rods. Fold over and staple one end of the drinking straw, slide the element rods into the straw, and then fold over and staple the second end of the straw closed. This keeps the element rods together and prevents them from poking their way out of a plastic bag that may be used to contain all of the parts that comprise an antenna kit.